CHENNAI
MATHEMATICAL
INSTITUTE

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MSc/PhD Computer Science Entrance Examination

Pattern and Syllabus

Exam Pattern:

The examination will have three parts: Part A, Part B and Part C.

Part A (30 marks): There will be 10 multiple choice questions, each carrying 3 marks.

Part B (30 marks): There will be 3 long-answer questions, each carrying 10 marks.

Part C (40 marks): There will be 8 long-answer questions, each carrying 10 marks. Students can choose any 4 out of these 8 questions. Students may attempt more than 4 questions; in such cases, the best 4 answers will be considered for the final score.

Syllabus:

Part A. Discrete mathematics, logical reasoning, elementary programming questions.

Part B. Discrete mathematics, logical reasoning, graphs.

Part C.

- 4 questions from

  Discrete mathematics, graphs, formal languages and automata theory, basic data structures and algorithms.

- 4 questions from

  Probability theory, calculus, linear algebra

Topics covered:

- **Discrete Mathematics**

  Elementary combinatorics, principle of induction, pigeon hole principle, permutations and combinations, basic finite set theory, functions and relations.

- **Logic**

  Boolean logic, truth tables, boolean circuits — and, or, not, nand gates.

- **Graphs**

  Basic definitions, trees, bipartite graphs, matchings in bipartite graphs, breadth first search, depth first search, minimum spanning trees, shortest paths.

- **Formal Languages and Automata Theory**

  Regular expressions, non-deterministic and deterministic finite automata, subset construction, regular languages, non-regularity (pumping lemma), context free grammars, basic ideas about computable and noncomputable functions.

- **Algorithms**

  O notation, recurrence relations, time complexity of algorithms, sorting and searching (bubble sort, quick sort, merge sort, heap sort).

- **Algorithmic Techniques**

  Dynamic programming, divide and conquer, greedy.

- **Data Structures**

  Lists, queues, stacks, binary search trees, heaps.

- **Probability Theory**

  Finite probability spaces, Bayes theorem, conditional probability, elementary distributions, expectation, variance, moments, Markov and Chebyshev bounds.

- **Calculus**

  Limits, continuity, differentiability, functions of several variables, derivatives and partial derivatives, integral calculus.

- **Linear Algebra**

  Vector spaces, linear operators, eigenvalues and eigenvectors, linear equations, determinants, linear independence, inner product spaces, symmetric and Hermitian matrices.

Suggested Reading Material:

1. Frank Harary: *Graph Theory*, Narosa.
2. John Hopcroft and Jeffrey D. Ullman: *Introduction to Automata, Languages and Computation*, Narosa.
3. Jon Kleinberg and Eva Tardos: *Algorithm Design*, Pearson.
4. C. Liu: *Elements of Discrete Mathematics*, Tata McGraw-Hill.
5. Michael Huth and Mark Ryan: *Logic in Computer Science*, Cambridge
6. Sheldon Axler: *Linear Algebra done right*, Springer.
7. Gilbert Strang: *Calculus*, Wellesley-Cambridge Press
8. Sheldon Ross: *A First Course in Probability*, Pearson